8.14 二体修正

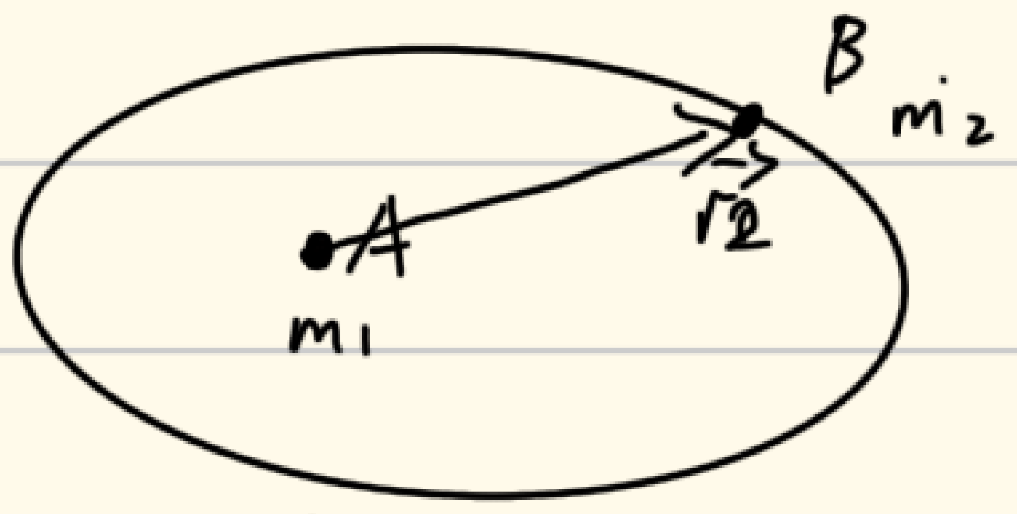
系统 · · 2个质点.

系统 物理 { 牛二 $m_1\ddot{\vec{r}}_1 = \vec{F}_{21} \quad m_2\ddot{\vec{r}}_2 = \vec{F}_{12}$ ①

结构力 $\vec{F}_{21} = \frac{Gm_1m_2}{|\vec{r}_1-\vec{r}_2|^3}(\vec{r}_1-\vec{r}_2)$ ②

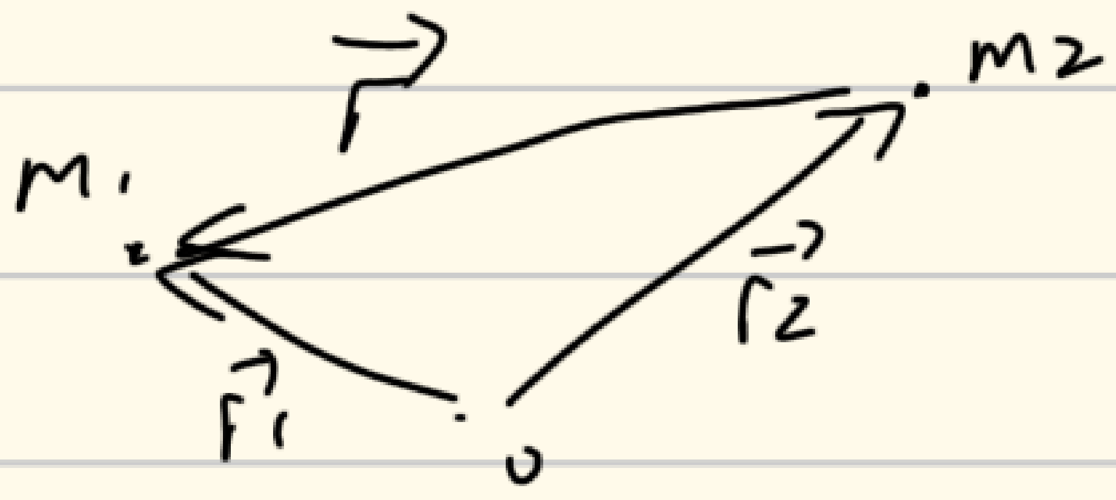
数学 + MATH

若一点定死



$$m_2\ddot{\vec{r}}_2 = -\frac{Gm_1m_2}{r_2^2}\hat{e}_{r_2} \quad *$$

若两点都自由



$$m_1\ddot{\vec{r}}_1 = \vec{F}_{21} \text{ ①} \quad m_2\ddot{\vec{r}}_2 = \vec{F}_{12} \text{ ②}$$

$$\vec{F}_{21} = \frac{Gm_1m_2}{|\vec{r}_1-\vec{r}_2|^3}(\vec{r}_1-\vec{r}_2)$$

$\frac{①}{m_1} - \frac{②}{m_2}$: 定义 $\vec{r}_1-\vec{r}_2 = \vec{r}$

$$\ddot{\vec{r}} = \frac{\vec{F}_{21}}{m_1} - \frac{\vec{F}_{12}}{m_2}$$

$$\ddot{\vec{r}} = -\vec{F}_{12}\left(\frac{1}{m_1}+\frac{1}{m_2}\right)$$

$$\underbrace{\frac{m_1m_2}{m_1+m_2}}_{\mu}\ddot{\vec{r}} = -\vec{F}_{12} \quad *$$

物理部分结束

$$\begin{cases} m_2\ddot{\vec{r}}_2 = \vec{F}(\vec{r}_2) \longrightarrow \{Math1\} \ ✓ \\ \mu\ddot{\vec{r}} = \vec{F}(\vec{r}) \longrightarrow \{Math2\} \ ✓ \end{cases}$$

$\frac{Gm_1m_2}{r^2}\vec{e}_r \quad \frac{K}{r^2}\vec{e}_r$

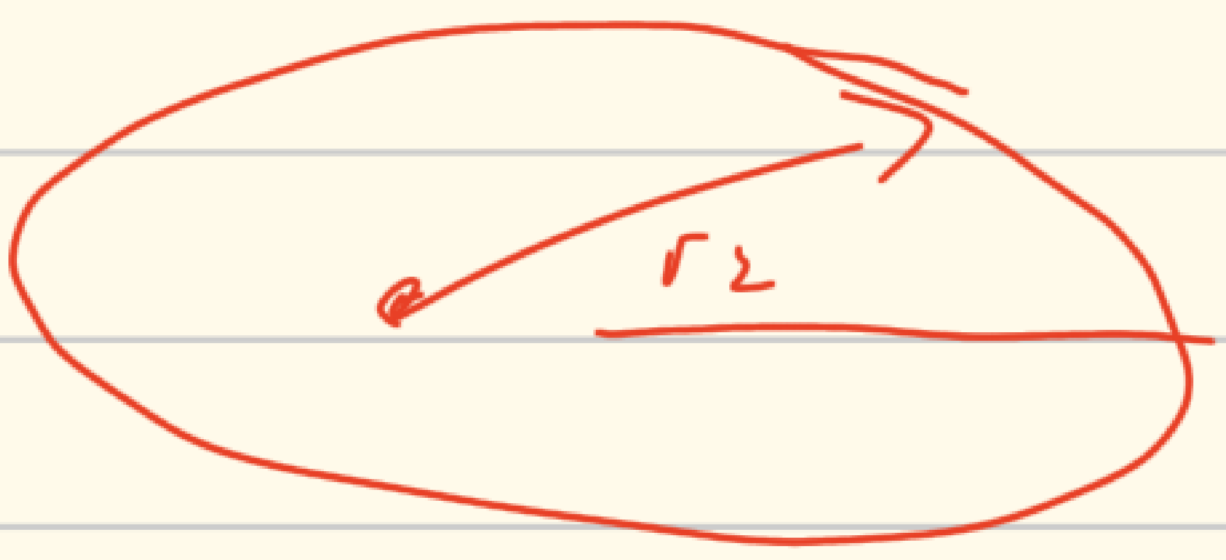
All m_2 in {Math1} replace → μ $\overset{?}{=}$ {Math2}

$$\begin{cases} m_2\ddot{\vec{r}}_2 = -\frac{K}{r_2^2}\vec{e}_{r_2} \\ \mu\ddot{\vec{r}} = -\frac{K}{r^2}\vec{e}_r \end{cases}$$

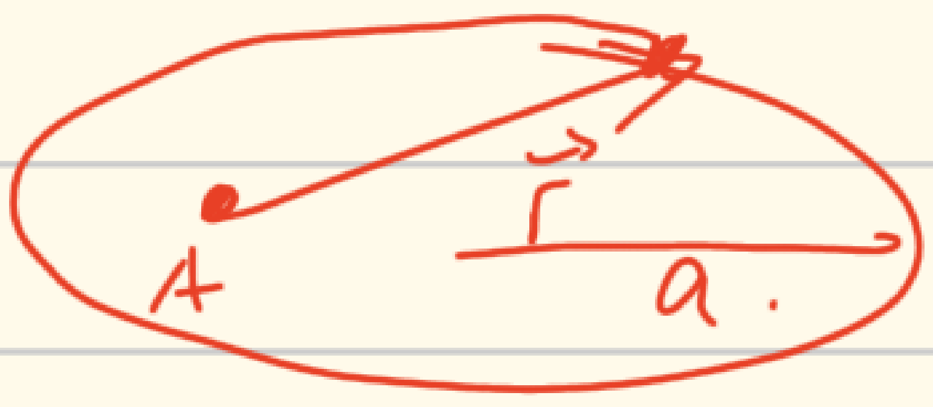
{Math1}

{Math2}

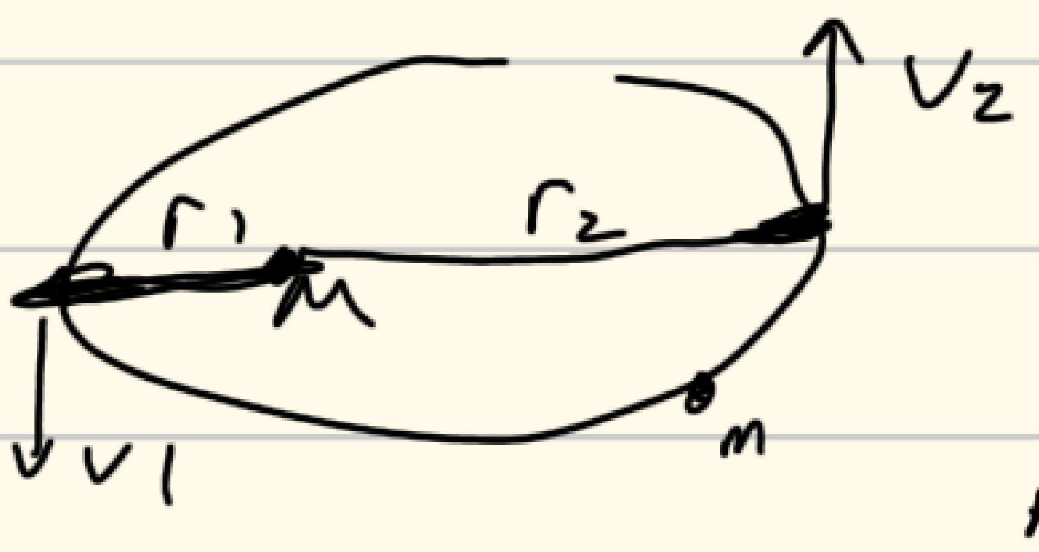
eg. 二体 E 单体: $-\frac{GMm}{2a}$



$\hookrightarrow -\frac{GMm}{2a} = E$



$$\begin{cases} \frac{1}{2}mV_1^2 - \frac{GMm}{r_1} = E \\ \frac{1}{2}mV_2^2 - \frac{GMm}{r_2} = E \end{cases}$$



$mr_1V_1 = L = mr_2V_2$

$$\Rightarrow \begin{cases} \frac{L^2}{2m}\frac{1}{r_1^2} - \frac{GMm}{r_1} = E \\ \frac{L^2}{2m}\frac{1}{r_2^2} - \frac{GMm}{r_2} = E \end{cases}$$

r_1, r_2 是方程 $\frac{L^2}{2m}\cancel{\ } - \frac{GMm\,r}{\cancel{r}} = Er^2$ 的两根

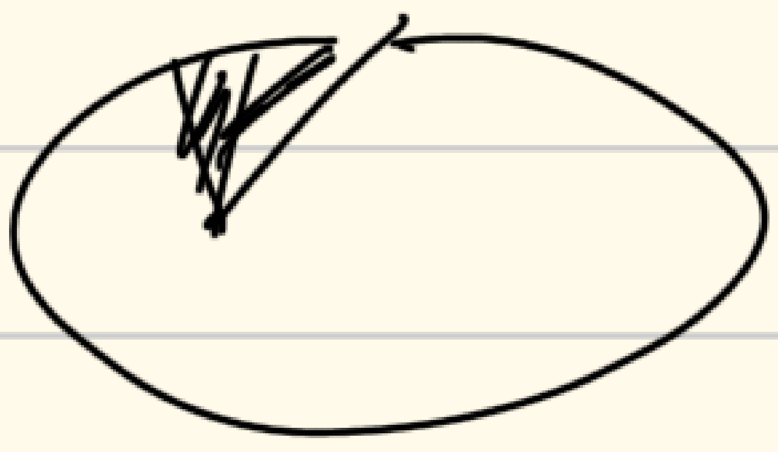
$r_1 + r_2 = -\frac{GMm}{E}$

$2a =$

$E = -\frac{GMm}{2a}$

eg2. $\frac{a^3}{T^2} = \frac{GM\cancel{m}}{4\pi^2\cancel{m}} = \frac{G(M+m)}{4\pi^2}$ $\quad \overset{K}{\frac{GMm}{4\pi^2 m}}$ $\quad -\frac{K}{r^2}\vec{e_r}$

$T = \frac{S}{\dot{S}} = \frac{\pi ab}{\dot{S}}$



$\dot{S} = \frac{1}{2}\vec{r}\times\vec{v} = \frac{L}{2m}$

$T = \frac{2m\pi ab}{L}$

$b = \sqrt{\frac{b^2}{a}\cdot a} = \sqrt{p\cdot a} = \sqrt{\frac{L^2}{\underset{\frac{GMm\cdot m}{K}}{GMm^2}}a}$

$T = \frac{2\pi ma}{L}\sqrt{\frac{L^2}{GMm^2}a}$ ①

$h^2u^2\left(\frac{\partial^2 u}{\partial\theta^2} + u\right) = -\frac{F}{m}$ $\qquad F = -\frac{GMm}{r^2} = -\frac{K}{r^2}\hat{e}_r$

$h^2u^2\left(\frac{\partial^2 u}{\partial\theta^2} + u\right) = \frac{K}{mr^2}$

$\left(\frac{\partial^2 u}{\partial\theta^2} + u\right) = \frac{K}{mh^2}$

$u = A\cos(\theta) + \frac{Km}{L^2}$

$\downarrow$

$\frac{1}{r} =$

$r = \frac{1}{\frac{Km}{L^2} + A\cos\theta}$

$r = \frac{\frac{L^2}{Km}}{1 + e\cos\theta}$

$\rightarrow p = \frac{L^2}{\underset{GMm}{K}m}$

$$T = \frac{2\pi m a}{L}\sqrt{\frac{L^2}{GMm^2}a}$$

$$T = \frac{2m\pi a}{L}\sqrt{\frac{L^2}{Km}a}$$

$$T^2 = \frac{4m\pi^2 a^3}{K}$$

$$\frac{a^3}{T^2} = \frac{K}{4\pi^2 \boxed{m}}\ \mu$$

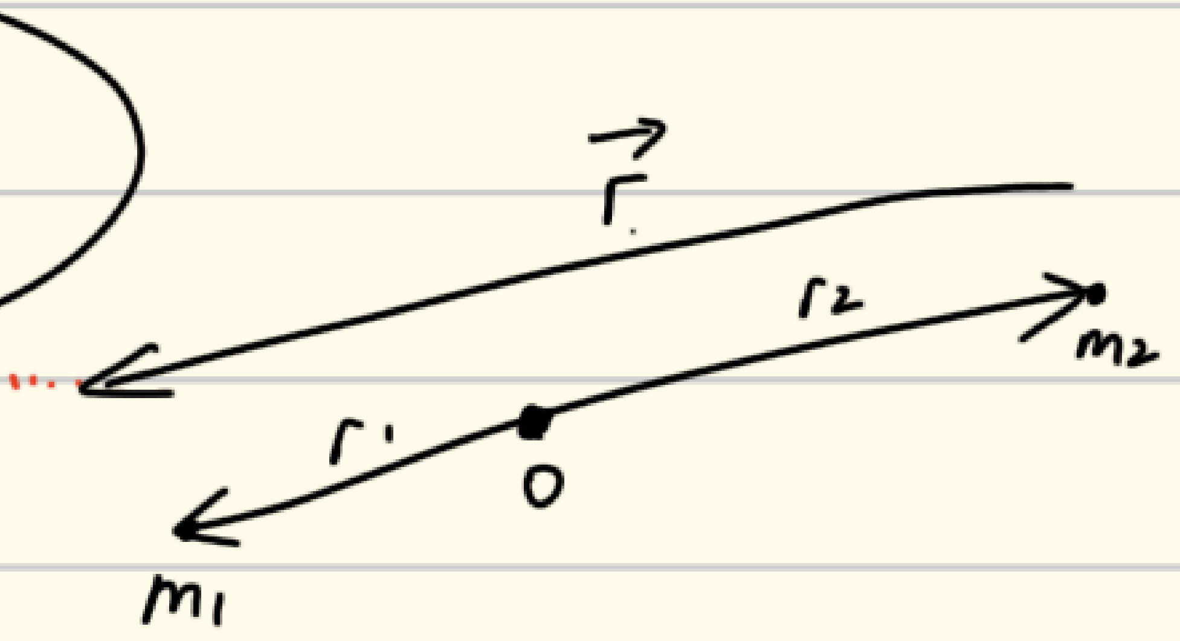
$$\frac{a^3}{T^2} = \frac{K}{4\pi^2\mu} = \frac{GMm}{4\pi^2\frac{Mm}{M+m}} = \frac{G\boxed{(M+m)}}{4\pi^2}$$

eg3.

$$e = \sqrt{1+\frac{2EL^2}{G^2M^2m^3}} \qquad p = \frac{L^2}{GMm^2}$$

$$\begin{cases} E = \frac{1}{2}m_2\dot{\vec{r}}_2^{\,2} - \frac{\overset{K}{\boxed{Gm_1m_2}}}{r_2} \\ L = m_2\vec{r}_2\times\dot{\vec{r}}_2 \end{cases}$$

$$\Rightarrow \boxed{\begin{cases} E = \frac{1}{2}\mu\dot{\vec{r}}^{\,2} - \frac{k}{r} \\ L = \mu\vec{r}\times\dot{\vec{r}} \end{cases}}$$



$$\vec{r}_1 = \frac{m_2}{m_1+m_2}\vec{r}$$

$$\vec{r}_2 = -\frac{m_1}{m_1+m_2}\vec{r}$$

$$\vec{L}_1 = m_1\vec{r}_1\times\dot{\vec{r}}_1 = \frac{m_2^2m_1}{(m_1+m_2)^2}\vec{r}\times\dot{\vec{r}}$$

$$\vec{L}_2 = m_2\vec{r}_2\times\dot{\vec{r}}_2 = \frac{m_1^2m_2}{(m_1+m_2)^2}\vec{r}\times\dot{\vec{r}}$$

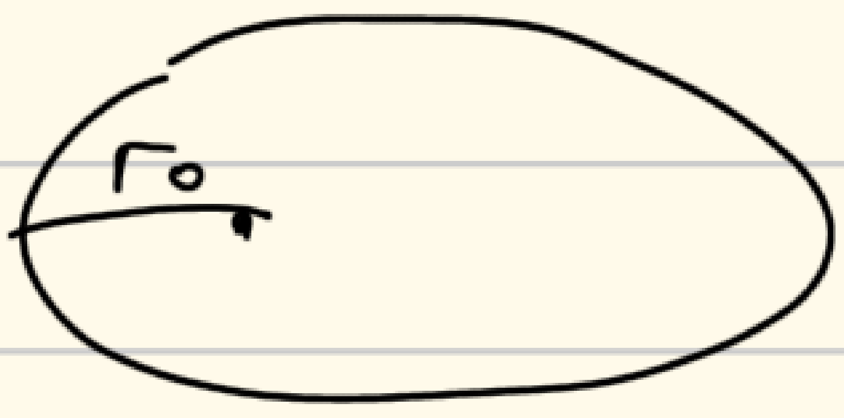
$$\vec{L} = \vec{L}_1+\vec{L}_2 = \frac{m_1m_2\cancel{(m_1+m_2)}}{(m_1+m_2)^{\cancel{2}}}$$

$$= \mu\vec{r}\times\dot{\vec{r}}$$

eg4

$$e = \sqrt{1+\frac{2\boxed{E}L^2}{\underset{K^2m.}{\boxed{G^2M^2m^3}}}} \ \overset{化简}{=} \ \sqrt{1+\frac{2\boxed{E'}L'^2}{\underset{K^2}{\boxed{G^2M^2m^2}}\mu}}$$

$$r = \frac{\boxed{p}}{1+e\cos\theta}$$



$$2a = -\frac{\overset{K}{\boxed{GMm}}}{E}$$

$$\frac{1}{a} = -\frac{2E}{K}$$

$$r_0 = \frac{p}{1+e} = a-c = a(1-e)$$

$$1-e^2 = \frac{p}{a} = -\frac{L^2}{Km}\frac{2E}{K}$$

$$1-e^2 = -\frac{2EL^2}{K^2m}$$

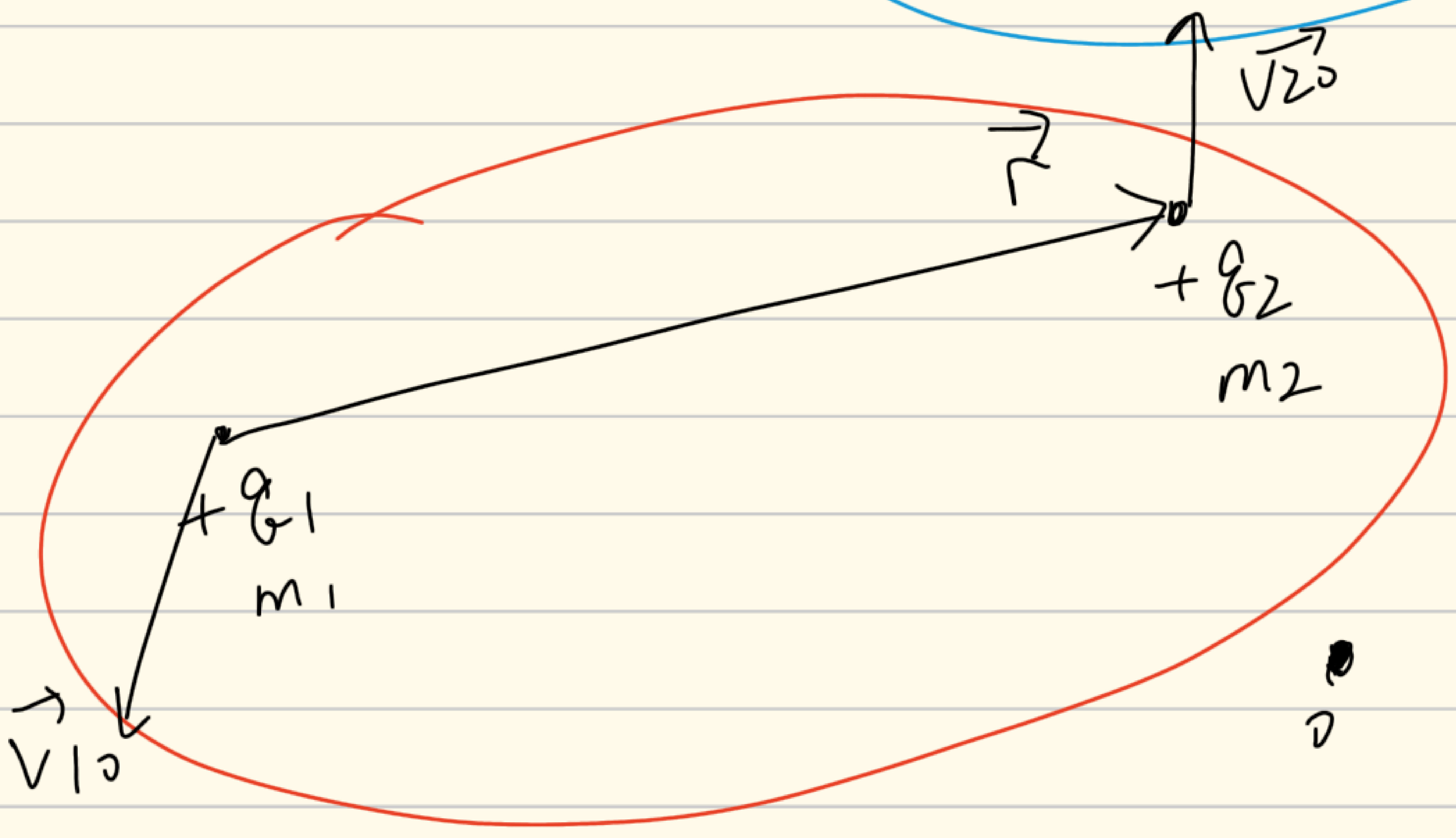
$$e = \sqrt{1+\frac{2EL^2}{K^2\boxed{m}}} \rightarrow \mu$$

eg4.

$\frac{GMm}{r^2} \vec{e_r}$

$$\vec{F} = \frac{K}{r^2} \hat{e_r}$$

$\vec{v_{10}}, \vec{v_{20}}, \vec{r_0}$



$\vec{r}(\theta)$

$\frac{P}{1-e\cos\theta}$

$\vec{r}(\theta)$

L' E'

$$\mu = \frac{m_1 m_2}{m_1+m_2}$$

$$L' = \left|\mu \vec{r_0} \times (\vec{v_{20}} - \vec{v_{10}})\right|$$

$$E' = \frac{1}{2}\mu(\vec{v_{20}} - \vec{v_{10}})^2 + \frac{q_1 q_2}{4\pi\varepsilon_0 r_0}$$

$$r = \frac{\dfrac{L'^2}{\frac{1}{4\pi\varepsilon_0} q_1 q_2 \mu}}{1 - \sqrt{1 + \dfrac{2E'L'^2}{(\frac{1}{4\pi\varepsilon_0})^2 q_1 q_2 \mu}}\cos\theta}$$

总结

二体修正

① $-\frac{K}{r^2}$ ✓

② E. $\vec{L}$.

③

μ